

Web Development

064

Building the websites and applications we use every day

Web Development involves building and maintaining websites and web applications, covering everything from visual design to backend server logic.

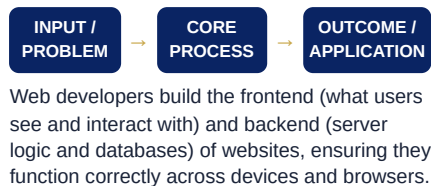
● DEFINITION

The work involved in developing a website or web application for the internet or a private network (intranet).

● WHY IT MATTERS

Websites and web applications are the primary way businesses and individuals interact online, making web development a foundational digital skill.

● WORKING PRINCIPLE



● QUICK FACTS

- ▶ The first website ever published is still online today at CERN's original URL
- ▶ There are now well over a billion websites on the internet

● KEY TERMS

Frontend Backend API
Responsive design

● CORE CONCEPTS

- ▶ Frontend development (HTML, CSS, JavaScript)
- ▶ Backend development and APIs
- ▶ Responsive design
- ▶ Web performance optimization

● APPLICATIONS

- ▶ E-commerce website development
- ▶ Web application development
- ▶ Content management systems
- ▶ Progressive web apps

● REAL-LIFE EXAMPLES

- ▶ Online shopping websites
- ▶ Social media platforms
- ▶ Online banking portals

● ADVANTAGES

- ▶ High demand with accessible entry paths
- ▶ Immediate, visible results of work
- ▶ Flexible freelance and remote opportunities

● LIMITATIONS

- ▶ Rapidly changing frameworks and tools
- ▶ Cross-browser compatibility challenges
- ▶ Security vulnerabilities require constant vigilance

CAREER OPPORTUNITIES

Frontend developer, Backend developer, Full-stack developer, Web application architect

RESEARCH AREAS

Web performance optimization, Progressive web applications, Web accessibility

INDUSTRY APPLICATIONS

E-commerce, Media and entertainment, Finance, Every business with an online presence

FUTURE SCOPE

AI-assisted development tools and increasingly powerful web frameworks are making sophisticated web applications faster to build.

● CURRENT TRENDS

Growing adoption of AI-powered code generation tools that speed up building web interfaces and features.

● SUMMARY

Web development builds the websites and applications that power online business and communication, combining visual design with backend logic.